Customer Churn Analysis

End-to-End Machine Learning Project

Project Title	Customer Churn Analysis & Prediction
Domain	Telecom / SaaS Analytics
Tech Stack	Python, SQL, scikit-learn, XGBoost, Streamlit, FastAPI
Database	SQLite (customer_churn_db)
Dataset Size	5,000 customers
Churn Rate	35.0%
Best Model	Logistic Regression
Best ROC-AUC	0.7078
Report Date	April 15, 2026

1. Executive Summary

This project delivers a complete, production-ready customer churn analysis pipeline that combines SQL-based analytics with machine learning to predict which customers are most likely to leave a subscription service. It covers the full data science lifecycle: synthetic data generation, database design, exploratory analysis, feature engineering, model training and evaluation, interactive dashboards, and REST API deployment.

The pipeline was trained on **5,000 customer records** with an overall churn rate of **35.0%**. Three models were compared, with **Logistic Regression** achieving the best F1 score of **0.454** and ROC-AUC of **0.708**.

Key Deliverables

- SQLite database with schema, 5,000 records, and 10 analytical queries
- Three trained ML models (Logistic Regression, Random Forest, XGBoost)
- Eight exploratory and evaluation visualizations
- Interactive Streamlit dashboard with 5 pages
- FastAPI REST endpoint with Swagger documentation
- Docker + docker-compose deployment configuration
- Cloud deployment blueprints (Render, Heroku)
- Reusable predict_churn() function and saved preprocessing artifacts

2. Problem Statement

Customer churn — the rate at which customers stop doing business with a company — is one of the most critical metrics for subscription-based businesses. Acquiring a new customer typically costs 5–7× more than retaining an existing one, so even small reductions in churn translate directly into significant revenue gains.

The goal of this project is to build a system that can (1) analyze historical customer data to identify the key drivers of churn, and (2) predict in real time which customers are at highest risk of leaving, so the business can intervene with targeted retention offers before the customer actually churns.

3. Project Architecture

The project follows a modular architecture that separates data, SQL, source code, and application layers for clean development and deployment.

```
customer_churn_project/
■■■ data/              → SQLite database
■■■ sql/
■   ■■■ schema.sql     → Table definitions & indexes
```

```
■   ■■■ queries.sql    → 10 analytical queries
■■■ src/
■   ■■■ data_loader.py → Synthetic generator + DB operations
■   ■■■ preprocessing.py → Cleaning, encoding, scaling
■   ■■■ model.py       → Training, tuning, prediction
■   ■■■ evaluate.py    → Metrics and evaluation plots
■   ■■■ visualize.py   → EDA visualizations
■■■ app/
■   ■■■ streamlit_app.py → Interactive dashboard
■   ■■■ api.py         → FastAPI endpoint
■■■ models/            → Pickled models + artifacts
■■■ plots/             → Generated visualizations
■■■ deploy/            → Render, Heroku configs
■■■ Dockerfile
■■■ docker-compose.yml
■■■ main.py            → End-to-end pipeline
■■■ requirements.txt
```

4. Dataset Description

A realistic synthetic telecom dataset was generated with probabilistic churn labels that depend on contract type, tenure, monthly charges, payment method, and internet service — mirroring real-world patterns.

Column	Type	Description
customer_id	TEXT	Unique customer identifier
gender	TEXT	Male / Female
age	INTEGER	Customer age (18–75)
tenure	INTEGER	Months as a customer (0–72)
monthly_charges	REAL	Current monthly bill ($)
total_charges	REAL	Lifetime charges ($)
contract_type	TEXT	Month-to-month / One year / Two year
payment_method	TEXT	Electronic check / Mailed check / Bank / Credit
internet_service	TEXT	DSL / Fiber optic / No
churn	TEXT	Target variable: Yes / No

Dataset Statistics

Metric	Value
Total Records	5,000
Features	8
Churn Rate	35.00%
Avg Monthly Charges	$60.65
Avg Tenure	35.6 months
Avg Age	46.3 years

5. SQL-Based Exploratory Analysis

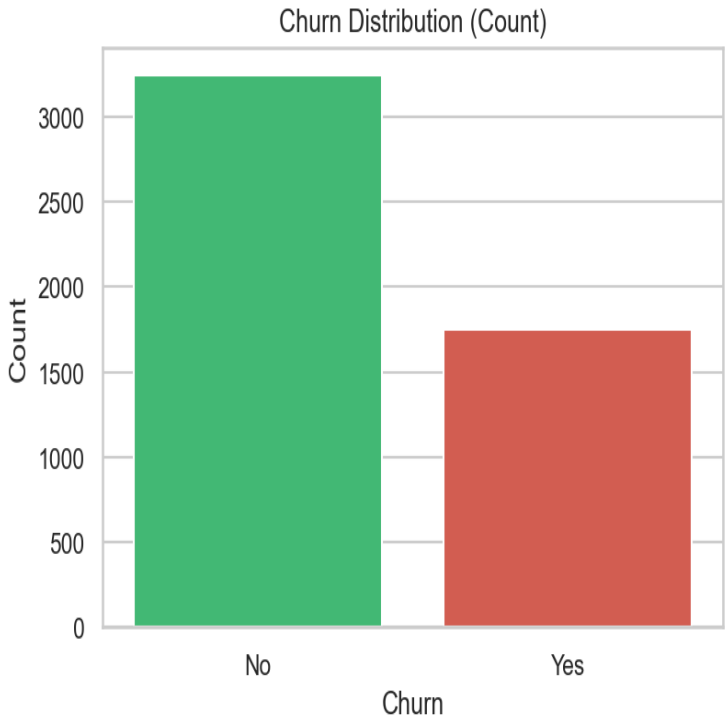
Ten analytical SQL queries were written to explore churn patterns directly in the database. Key findings from these queries are shown below.

5.1 Contract Type Churn Rate

5.2 Revenue Impact of Churn

6. Exploratory Data Visualizations

6.1 Churn Distribution



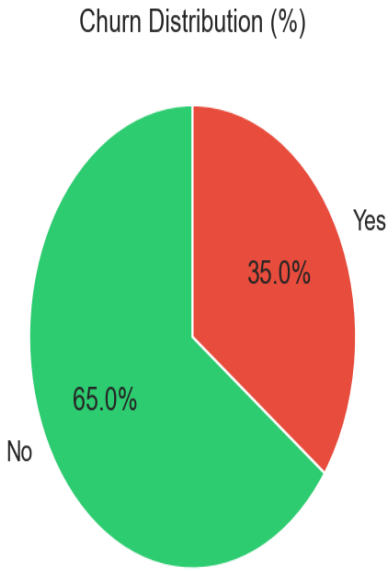
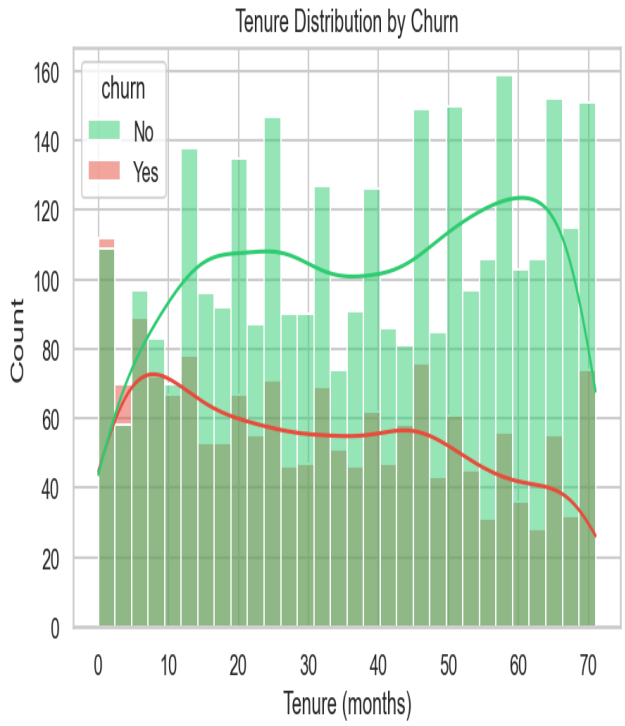


Figure: Overall churn class balance

6.2 Tenure vs Churn



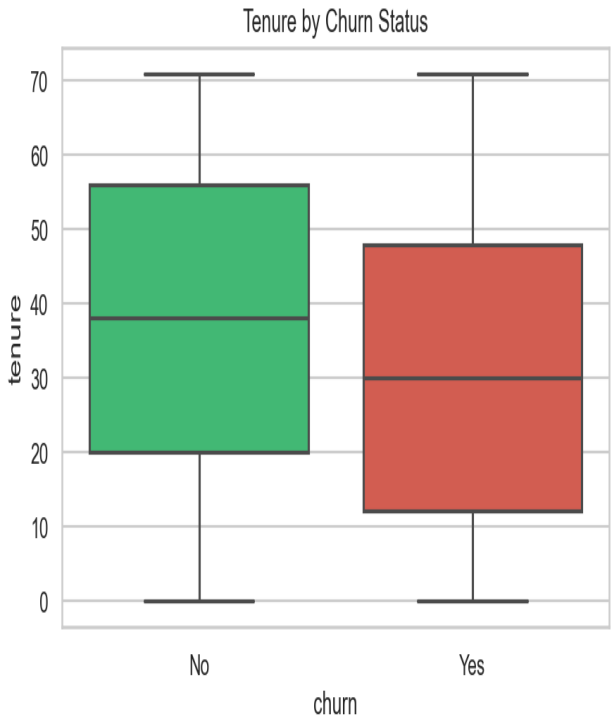


Figure: Customers with low tenure churn more often

6.3 Monthly Charges vs Churn

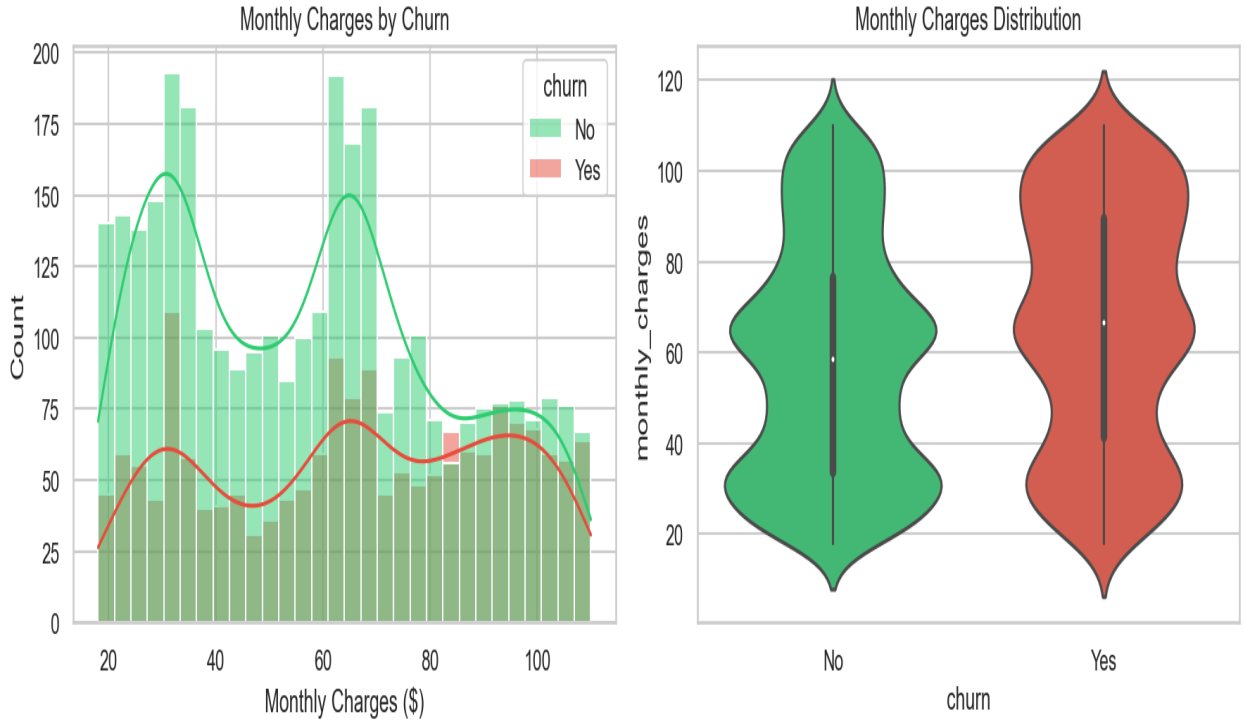


Figure: Higher monthly charges correlate with churn

6.4 Correlation Heatmap

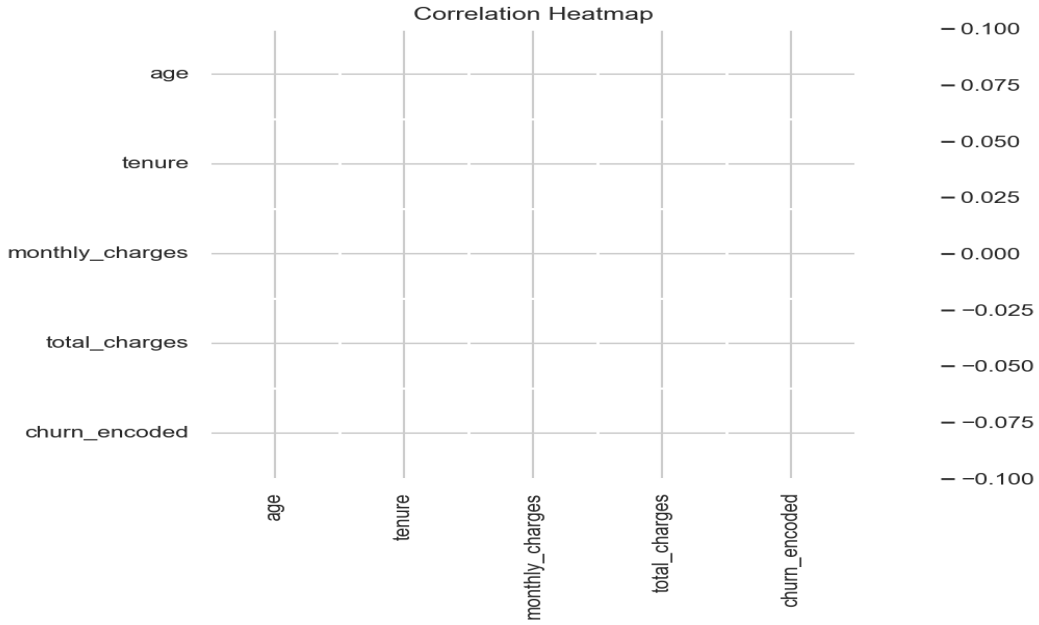


Figure: Numerical feature correlations

6.5 Contract Type Analysis

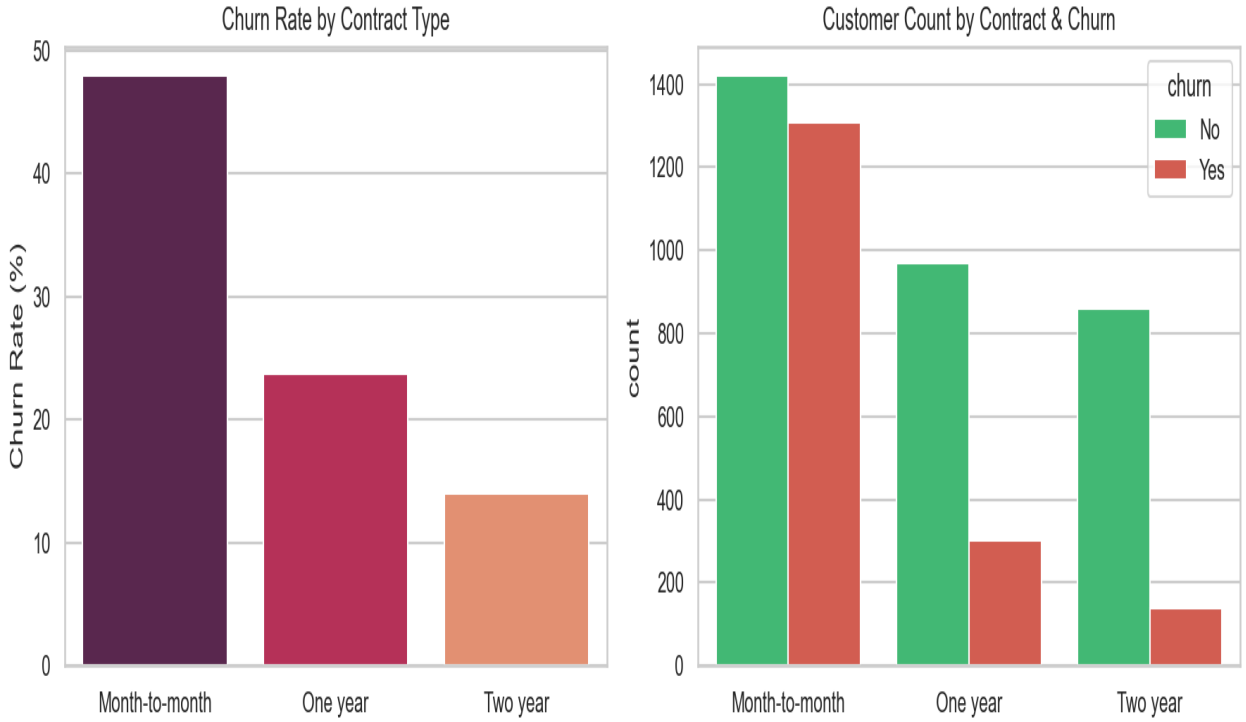


Figure: Month-to-month contracts drive most churn

7. Machine Learning Pipeline

7.1 Preprocessing

- **Cleaning:** Handle missing values, remove duplicates, enforce data types.
- **Encoding:** LabelEncoder for categorical features (gender, contract, payment, internet).
- **Scaling:** StandardScaler on numerical features (age, tenure, charges).
- **Split:** Stratified 80/20 train/test split to preserve churn ratio.

7.2 Models Trained

Model	Description	Use Case
Logistic Regression	Linear probabilistic classifier	Interpretable baseline
Random Forest	Bagged tree ensemble	Non-linear patterns
XGBoost	Gradient-boosted trees	High accuracy

7.3 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.685	0.577	0.374	0.454	0.708
XGBoost	0.658	0.516	0.377	0.436	0.682
Random Forest	0.662	0.525	0.354	0.423	0.690

7.4 Confusion Matrices

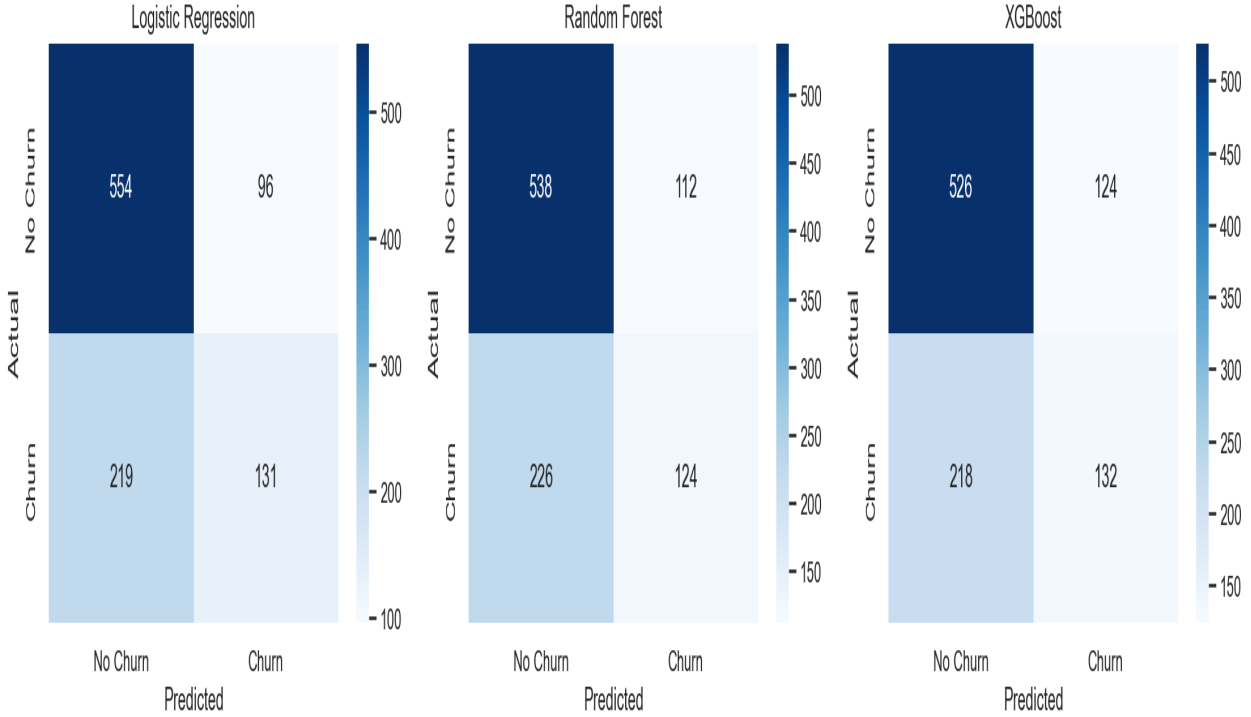


Figure: True vs predicted labels for all three models

7.5 ROC Curves

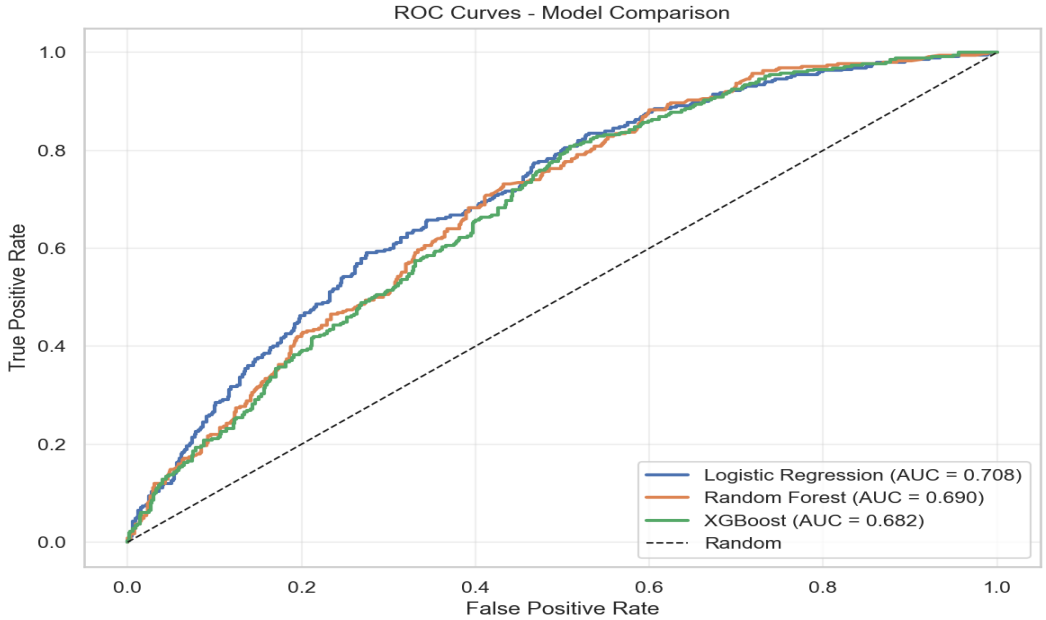


Figure: ROC-AUC comparison across all models

7.6 Feature Importance

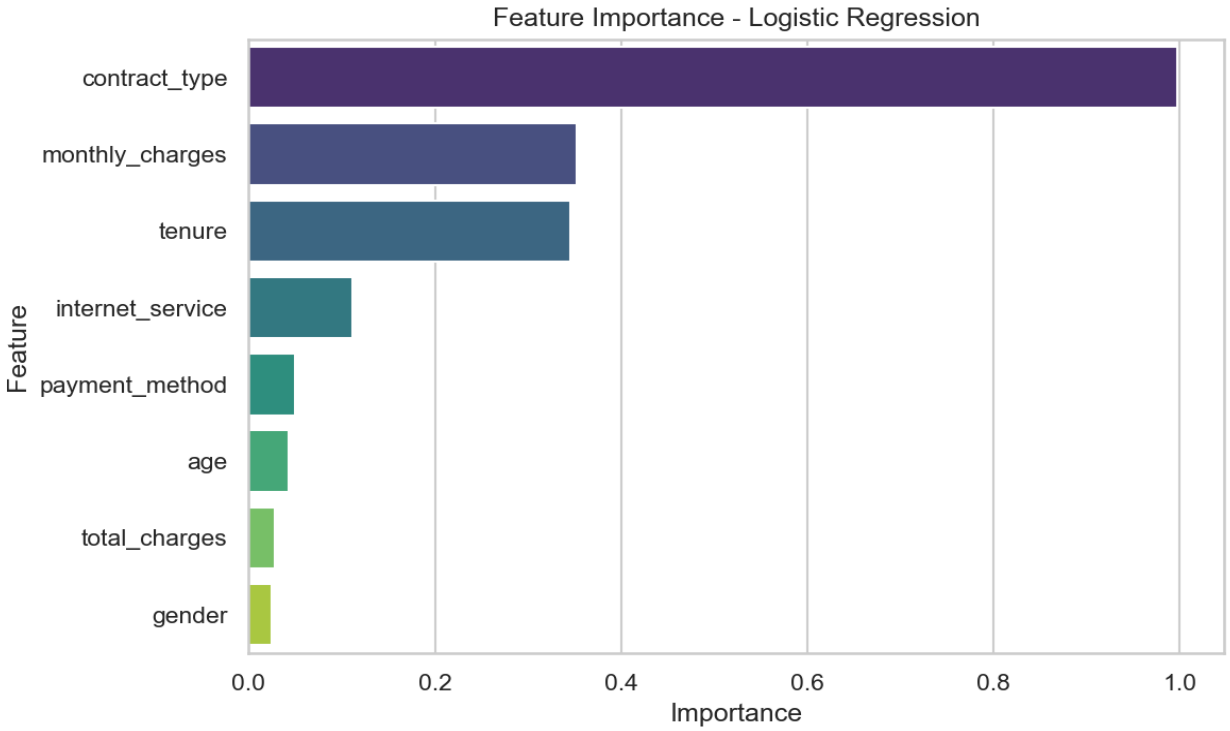


Figure: Top churn drivers (Logistic Regression)

Rank	Feature	Importance
1	contract_type	0.9976
2	monthly_charges	0.3525
3	tenure	0.3455
4	internet_service	0.1110
5	payment_method	0.0495

8. Key Business Insights

- **Contract type is the strongest churn predictor.** Month-to-month customers churn at dramatically higher rates than one-year or two-year contracts. Incentivizing longer contracts is the highest-ROI retention lever.
- **Low tenure = high churn risk.** New customers (< 12 months) are far more likely to churn. Early-lifecycle engagement programs are critical.
- **High monthly charges increase churn.** Customers paying above $80/month show elevated churn, especially on fiber-optic plans. Consider loyalty discounts.
- **Payment method matters.** Electronic check users churn more than those on automatic bank transfer or credit card — auto-pay reduces friction.
- **Internet service type is a factor.** Fiber optic customers churn more often than DSL or no-internet customers, suggesting quality or pricing issues with the premium tier.

9. Deployment

The project ships with multiple deployment options covering local, containerized, and cloud environments.

9.1 Local Development

```
pip install -r requirements.txt
python main.py
streamlit run app/streamlit_app.py
uvicorn app.api:app --reload
```

9.2 Docker

```
docker-compose up --build
# Dashboard → http://localhost:8501
# API → http://localhost:8000/docs
```

9.3 Cloud Deployment

Render: one-click deploy via deploy/render.yaml blueprint
Heroku: deploy/Procfile for web + release steps
Streamlit Cloud: point to app/streamlit_app.py in a GitHub repo

10. REST API Reference

Method	Endpoint	Description
GET	/	Root — API info
GET	/health	Health check

Method	Endpoint	Description
GET	/docs	Swagger documentation UI
POST	/predict	Predict churn for a single customer
GET	/analytics/churn-rate	Overall churn statistics

Example Request

```
POST /predict
Content-Type: application/json

{
  "gender": "Female",
  "age": 25,
  "tenure": 2,
  "monthly_charges": 95.0,
  "total_charges": 190.0,
  "contract_type": "Month-to-month",
  "payment_method": "Electronic check",
  "internet_service": "Fiber optic"
}
```

Example Response

```
{
  "prediction": "Yes",
  "churn_probability": 0.7009,
  "retention_probability": 0.2991
}
```

11. Technology Stack

Layer	Technology	Purpose
Language	Python 3.11	Primary runtime
Database	SQLite	Customer data storage
Data	pandas, numpy	Data manipulation
ML	scikit-learn	Preprocessing, LR, RF
ML	XGBoost	Gradient boosting model
Viz	matplotlib, seaborn	EDA and evaluation plots
Dashboard	Streamlit	Interactive web UI
API	FastAPI + Pydantic	REST endpoint + validation
Server	Uvicorn	ASGI server
Container	Docker + Compose	Reproducible deployment
Report	ReportLab	PDF generation

12. Future Improvements

- Replace synthetic data with a real telecom dataset (e.g., IBM Telco Churn)
- Add SHAP for per-prediction explainability in the dashboard
- Implement model monitoring and data drift detection
- Add A/B testing framework for retention campaigns
- Migrate from SQLite to PostgreSQL for production scale
- Add CI/CD pipeline (GitHub Actions) for automated retraining
- Implement authentication for the API (JWT / API keys)
- Add unit tests and integration tests with pytest

13. Conclusion

This project demonstrates a complete end-to-end machine learning workflow for customer churn prediction — from raw data generation through SQL analytics, feature engineering, model training and evaluation, all the way to production-grade deployment via Streamlit, FastAPI, and Docker. The modular architecture makes it straightforward to swap in real data, add new models, or extend the dashboard with additional analytics. The trained models, together with the SQL insights, form a

foundation that a business team can use today to identify at-risk customers and drive targeted retention campaigns.

— End of Report —